



ANNEALING AS ROBUST BAYESIAN INFERENCE

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- Given a likelihood $L(y | x)$ and prior $p(x)$, the posterior $p(x | y)$

$$p(x | y) = \frac{L(x | y)p(x)}{p(y)}$$

How are we that our model is correct? Are we robust to the likelihood, outliers, and the prior,

► Where L_β interpolates between 1 and L .

Are we robust to the likelihood, outliers, and the prior,

- We can measure the sensitivity of our inference by annealing the likelihood

$$p_{\beta}(x | y) = \frac{L_{\beta}(x | y)p(x)}{p_{\beta}(y)}$$

► Hence p_β interpolates between the prior and posterior



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- ▶ Given a likelihood $L(y | x)$ and prior $p(x)$, the posterior $p(x | y)$

$$p(x | y) = \frac{L(x | y)p(x)}{p(y)}$$

- ▶ How sure are we that our model is correct? Are we robust to the likelihood, outliers, and the prior,
- ▶ Are we robust to the likelihood, outliers, and the prior,
- ▶ We can measure the sensitivity of our inference by annealing the likelihood

$$p_{\beta}(x | y) = \frac{L_{\beta}(x | y)p(x)}{p_{\beta}(y)}$$

- ▶ Where L_{β} interpolates between 1 and L .
- ▶ Hence p_{β} interpolates between the prior and posterior
- ▶ Can see the influence of the data/model

LINEAR PATH AS POWER POSTERIOR